Project 1 — Classification of Abnormal Electric-Vehicle Charging Sessions

1. Context

Electric-vehicle charging infrastructures generate large volumes of operational data. Identifying charging sessions at increased risk of **abnormal termination** can support preventive monitoring, charger maintenance, resource allocation, and service-quality improvement.

In this project, you will develop and evaluate a classification solution for predicting whether an electric-vehicle charging session will terminate abnormally.

Each group must design, justify, implement, and critically evaluate its own modelling solution.

2. Prediction objective

The objective is to predict the binary target:

- is_Abnormal = 1: abnormal charging session;
- is_Abnormal = 0: normal charging session.

The prediction must represent a decision made **when a charging session starts**. Consequently, only information that would reasonably be available at that time may be used as a predictor.

The project should answer the following general question:

> To what extent can contextual, operational, and historical information available at the beginning of a charging session be used to **predict abnormal termination of a charging session**?

3. Available data

The dataset provided for this project is **an educational adaptation** of the China EV Charging Dataset introduced by Zhang et al. (2025). The original dataset contains 441,077 charging transactions recorded over two years at 13 public charging stations located in three districts of Jiaxing, China.

The project dataset retains variables describing the complete charging transaction, including information recorded at the beginning, during, and after the charging session. It also includes electricity-tariff information and precomputed historical attributes describing previous activity and abnormal sessions associated with charging posts and users.

The available variables are organised into the following groups.

Session identification and location

- UserID: identifier of the user associated with the transaction;
- Charging Post ID: identifier of the charging post;

- Location Information: charging-station location;
- District Name: district in which the station is located.

Charging-transaction information

- Order creation time: date and time of the order charge
- Transaction power/kWh: energy associated with the charging transaction;
- Electricity cost/Yuan: cost of the electricity consumed;
- Service charge/Yuan: service fee applied to the transaction;
- Transaction Amount/Yuan: total transaction amount;
- Actual Payment/Yuan: amount effectively paid;
- Start Time: date and time at which the charging session started;
- End Time: date and time at which the charging session ended;
- Payment time: date and time at which payment was recorded;
- End cause: reason why the charging session ended.

Weather conditions

- Temperature(℃): recorded temperature;
- Relative Humidity(%): recorded relative humidity;
- Precipitation(mm): recorded precipitation.

Consider weather variables as estimates or forecasts available at the beginning of the charging session.

Electricity-tariff information

- electricity_price: electricity price applicable at the session start time;
- tariff_period: tariff interval determined by the session start time.

The dataset contains precomputed attributes describing previous activity at the same charging post and at the same user, including:

Historical charging-post information

- numbers of previous completed sessions over different temporal windows;
- numbers and smoothed rates of previous completed abnormal sessions;
- time since the previous completed session;
- time since the previous completed abnormal session;
- indicators identifying posts without previous completed-session or abnormal-session history.

Historical user/account information

- is new user
- numbers of previous completed sessions over different temporal windows;
- numbers and smoothed rates of previous completed abnormal sessions;
- time since the previous completed session;
- time since the previous completed abnormal session;
- indicators identifying new users/accounts and accounts without completed-session history;
- number of other sessions associated with the same account that were active when the current session started.

Earlier-started sessions that remained active at the prediction time were excluded from all outcome-derived historical attributes. Their outcomes were not yet available. Concurrent activity is represented separately through user_active_sessions_at_start, which uses only start and end timestamps available at the prediction time.

Historical attributes were constructed chronologically to prevent the current session or future sessions from contributing information to the current predictors. Students are not required to reconstruct these variables, but they must:

- understand their meaning;
- evaluate their contribution through feature-group ablation;
- consider whether they would be operationally available at prediction time;
- discuss their implications for privacy, generalisation, and cold-start cases.

4. Main tasks

Task 1 — Understand, audit, and prepare the data

Conduct a systematic and critical analysis of the supplied data before developing the classification models. The objective is to understand the dataset, identify factors that may affect modelling, and define a justified data-preparation strategy.

As guidance, the analysis may include aspects such as data types and variable meanings, missing values, duplicated or inconsistent observations, temporal coverage, target-class distribution, numerical distributions, potential outliers, categorical cardinality, rare categories, relationships between variables, temporal changes, and possible data-quality problems. These examples are not exhaustive.

Potential data-quality problems must be investigated before corrective action is taken. Decisions to retain, transform, impute, group, cap, or remove observations must be supported by evidence and by the meaning of the corresponding variables.

Explore the data using appropriate numerical summaries and graphical representations. Figures should be selected to answer relevant analytical questions and must be interpreted in the report. A collection of plots without an explanation of their meaning, modelling implications, or relationship to the project objective will not be considered an adequate exploratory analysis.

The exploratory analysis should consider whether important properties of the data change over time. In particular, investigate whether session volume, class prevalence, predictor distributions, user or charging-post activity, and other relevant patterns remain stable throughout the observation period. Explain how any identified temporal changes may affect model development and evaluation.

Any preparation operation that learns information from the data—including imputation, scaling, category grouping, outlier thresholds, resampling, feature selection, or dimensionality reduction represent different methodological reasons and must be documented separately.

At the end of this task, provide a concise synthesis identifying:
- the most important characteristics and problems discovered in the data;
- the evidence supporting the adopted preparation decisions;
- the variables or variable groups retained for modelling;
- unresolved limitations or assumptions;
- the expected implications for model selection and evaluation.

The quality of this task will be assessed through the relevance and depth of the investigation, the justification of decisions, and the interpretation of evidence—not by the number of statistics, figures, or preprocessing operations presented.

Task 2 — Define the prediction and evaluation setting

Formally define the classification problem and the conditions under which the proposed solution will be evaluated.

Begin by specifying:
- the unit of observation;
- the positive and negative classes;
- how the target is derived from end_cause;
- the moment at which the prediction is intended to be made;
- the purpose for which the prediction could be used;
- the operational meaning of predicting a session as abnormal.

Determine which supplied variables would genuinely be available at the selected prediction time. Classify the variables into the following categories:
1. variables eligible for use as model predictors;
2. variables used exclusively to construct the target;
3. variables excluded because they are generated during or after the session;
4. variables excluded for another methodological reason.

Every exclusion must be justified.

Evaluation objectives and error consequences

Analyse the consequences of the two main classification errors:
- a false positive, in which a normal session is incorrectly flagged as abnormal;
- a false negative, in which an abnormal session is not detected.

Discuss which type of error is likely to be more important in the proposed application and why. Consider the actions that might follow an alert, such as automated monitoring, technical inspection, maintenance prioritisation, or user intervention. The relative importance of these errors should guide the selection of evaluation metrics and, later, the decision threshold.

Selection of evaluation metrics

Define an evaluation framework appropriate for the class distribution and operational objective. Select:

- a primary metric for model comparison and optimisation;
- complementary predictive metrics;
- operational metrics that describe the practical consequences of the predictions;
- measures for evaluating performance stability across temporal periods.

The selected metrics must be justified before model optimisation begins. The justification should explain what each metric measures, why it is relevant, and what limitations it has in this problem.

Beside model evaluation metrics you may additionally consider operational evaluation metrics:

- correct alerts per 1,000 sessions;
- false alerts per 1,000 sessions;
- missed abnormal sessions per 1,000 sessions.

This list provides guidance and is not a substitute for reasoned metric selection. Students may propose additional metrics when they are relevant to the defined operational setting.

Design a temporal evaluation strategy

The observations must not be randomly divided without considering time. Training observations must precede the observations on which the model is evaluated.

Design a temporal validation strategy that supports:

- model comparison;
- hyperparameter selection;
- decision-threshold selection;
- unbiased final evaluation.

An expanding-window or rolling-origin strategy is recommended for model development. A final **chronologically later test period must remain untouched** until all modelling decisions have been completed.

Explain and justify:

- the selected temporal boundaries;
- the number of validation folds;
- any differences in fold sizes;
- how preprocessing is fitted inside each training fold;
- how the final test set is protected from model-selection decisions.

Aggregation across temporal folds

Explain how performance will be summarised across temporal evaluation periods. The evaluation plan should consider:

- mean performance;
- variability across folds;
- differences in fold sizes;
- weighted, unweighted, or pooled results;
- possible deterioration or improvement over time.

A model should not be judged solely by its average score if its performance is unstable across temporal periods. At the end of this task, present a concise evaluation protocol specifying:

- the admissible predictor set;
- the variables excluded and the reasons for exclusion;
- the selected primary and complementary metrics;
- the operational measures;
- the baseline strategy;
- the method used to aggregate and compare results across temporal folds.

This protocol must be established before hyperparameter optimisation and final model selection.

Establish an appropriate baseline

Define the reference against which model performance will be assessed. This should include an appropriate no-skill or simple baseline and a correct interpretation of its expected performance under the selected metrics.

Task 3 — Individual and ensemble prediction models

Investigate the contribution of different feature groups
Perform an ablation study to determine which types of information contribute to predictive performance.

At a minimum, compare:

1. contextual and weather variables;
2. contextual, weather, and tariff variables;
3. the previous variables plus charging-post history;
4. the previous variables plus charging-post and user history.

The purpose is not merely to identify the combination with the highest score. You must analyse:

- whether the tariff provides information beyond time-of-day variables;
- how much charging-post history contributes;
- whether user history produces an additional benefit;
- whether the effect is consistent across temporal periods;
- whether particular feature groups introduce operational or ethical concerns.

Groups may propose additional leakage-safe features, provided their construction and availability at prediction time are clearly justified.

Develop individual classification models
Select and evaluate classifiers representing at least three different modelling principles:

- linear classifiers;
- probabilistic classifiers;
- instance-based methods;
- decision trees;
- kernel-based methods.

The choice of algorithms is the responsibility of each group. It must be justified in relation to:

- data types;
- class imbalance;
- model assumptions;
- interpretability;
- computational requirements;
- expected relationships between predictors and the target.

Using many algorithms without analysis is not considered preferable to a smaller, well-justified comparison.

Develop ensemble models

Evaluate at least three ensemble approaches representing different ensemble principles.
Possible approaches include:

- bagging;
- random forests;
- boosting;
- soft or hard voting;
- stacking.

If stacking is used, the predictions used to train the meta-model must be generated without leakage. Standard random internal cross-validation is not appropriate when it allows future observations to influence predictions for earlier observations.

Explain why each ensemble could reasonably improve upon its constituent models.

Task 4 — Optimise selected models

Select a **limited number** (two or three) of **promising models** for hyperparameter optimisation.

The search procedure must:

- operate only on training and validation data;
- respect temporal order;
- use an explicitly defined search space;
- optimise a metric appropriate for the positive class;
- report both the selected configuration and the search procedure.

Compare the optimised models with their initial configurations to determine whether optimisation produces a meaningful and stable improvement.

The default classification threshold of 0.50 may not provide the most suitable balance between precision and recall. For the strongest probabilistic model, investigate one threshold-selection criteria, such as:

- maximising F1;
- achieving a minimum recall;
- limiting the number of false alerts;
- minimising a cost function;
- satisfying an operational alert-rate constraint.

The threshold must be selected using validation data and then applied unchanged to the final test set. Discuss the operational consequences of the selected threshold.

Compare the model's behaviour at the default and selected thresholds. Analyse the resulting changes in:

- precision;
- recall;
- F1-score;
- alert rate;
- correct alerts per 1,000 sessions;
- false alerts per 1,000 sessions;
- missed abnormal sessions per 1,000 sessions.

The threshold must be selected using predictions generated from development data not used to fit the corresponding model. It must then be fixed and applied unchanged to the final test period.

Students must not select or modify the threshold after inspecting the final test results. If model outputs are interpreted as estimated probabilities rather than ranking scores, students should also consider whether probability calibration is necessary. Calibration analysis is optional unless probability estimates form part of the proposed operational use.

For the final model, interpret which variables or feature groups influence its predictions. The interpretation method must be appropriate for the selected model and its limitations must be acknowledged.

Each weekly submission should include the corresponding notebook and a concise report section explaining decisions, evidence, results, and conclusions—not merely describing the code.
The report should focus on reasoning, evidence, and critical analysis. It should not reproduce sections of source code.

References

Zhang, Y., Xu, T., Chen, T., et al. (2025). A high-resolution electric vehicle charging transaction dataset with multidimensional features in China. *Scientific Data, 12*, 643. https://doi.org/10.1038/s41597-025-04982-1